GROUP B9 · FINAL REVIEW · STUDY GUIDE FOR PERSON A

Modelling the Tractor-Trailer

Part 1 of the presentation: what kind of robot the truck is, which robotics principles govern it, and how we describe it with state variables and inputs. By the end you should be able to draw the whole model by hand and defend every symbol.

Your slides: 5–8 Your time: about 4 minutes Hand over to Person B (kinematics)

1. The whole project in one page
2. What your part must achieve
3. What a model is, and which kind we use
4. Two rigid bodies and one passive joint
5. Frames and rigid-body transforms
6. Configuration space
7. Rolling without slipping
8. Two inputs, four states: still controllable
9. Choosing the state and the inputs
10. How to draw the model by hand
11. What to say on each slide
12. Questions the professor may ask
13. Check yourself, with answers
14. One-page cheat sheet

1 The Whole Project in One Page

Read this first. Every presenter should be able to tell this story in one minute, because the professor may ask any of you how your part fits in.

The Problem

Cars such as the Mahindra BE 6 and XEV 9e can already park themselves (Autopark: 12 ultrasonic sensors, perpendicular, angled and parallel bays). A truck with a trailer cannot, and it is the vehicle that needs help most. The reason is physics: a tractor-trailer is two bodies joined at a hitch, and **when it reverses, the hitch angle grows on its own**. A 2° error becomes 60° after 28 m of reversing. If nobody corrects it, the trailer folds against the cab: a **jackknife**.

What We Built

A guidance system that takes a tractor-trailer from the yard gate into a chosen bay while the hitch angle never exceeds 60°. It has four layers:

Layer	What it does	Presented in
Vehicle model	Four numbers describe the rig; four equations say how they change for a given speed and steering	Part 1 (A), Part 2 (B)
Hybrid A* planner	Searches the yard once and returns a route the truck can actually drive, gate to bay	Part 3 (C)
NMPC controller	Every 0.25 s, predicts 9.6 s ahead and chooses the steering that follows the route with $\|\psi\| \le 60°$ as a hard limit	Part 3 (C)
Driver guidance	Turns the controller's output into gear, speed (km/h), steering-wheel turns and distance	Part 4 (D)

The Result

We tested 110 free bays (4 layouts × 6 random fills of parked trucks). All 110 were parked within tolerance, the hitch limit was never broken (worst case 50.1°), and a driver model without hitch awareness jackknifed on all 72 reversing manoeuvres.

The Chain of Ideas

THE STORY IN FOUR SENTENCES

(1) We model the truck as a wheeled robot with four state variables and two inputs. (2) Its kinematics show that reversing is unstable: the hitch angle grows exponentially. (3) So we plan a drivable route with Hybrid A* and follow it with an NMPC that looks ahead and treats the hitch limit as a hard constraint. (4) The result parks every test bay and tells a human driver exactly what to do.

Symbol	Meaning	Value
x_1, y_1	Position of the tractor's rear-axle centre	metres
$\theta_0,\ \theta_1$	Heading of the tractor, heading of the trailer (from the x axis, anticlockwise positive)	radians or degrees
$\psi = \theta_0 - \theta_1$	Hitch angle: how far the tractor is turned relative to the trailer	hard limit 60°
$v,\ \delta$	Speed (negative in reverse), front-wheel steering angle	$\|v\| \le 2$ m/s, $\|\delta\| \le 35^\circ$
$L_1,\ L_2,\ d$	Tractor wheelbase, hitch to trailer axle, hitch ahead of the rear axle	4 m, 8 m, 0.5 m

2 What Your Part Must Achieve

Your four minutes set up everything the others say. If the professor understands *what the four state variables are and why the truck is nonholonomic*, Parts 2 and 3 follow naturally. Aim to leave him with three ideas:

1. **The truck is a wheeled mobile robot**: two rigid bodies (tractor and trailer) joined by one passive revolute joint (the hitch).

2. **Its wheels roll without slipping**, which gives velocity constraints that cannot be integrated: the system is **nonholonomic**. It cannot move sideways, yet it can still reach any position and heading.
3. **We describe it with four states and two inputs**: $X = [x_1, y_1, \theta_0, \psi]^T$ and $U = [v, \delta]^T$. We choose the hitch angle ψ as a state because it is the quantity we must keep safe.

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3 What a Model Is, and Which Kind We Use

A **model** is two things: a short list of numbers that says where the robot is right now (the **state**), and a rule that says how those numbers change when we apply an **input**. Written generally:

$$\dot{X} = f(X, U)$$

The dot means "rate of change with time". Part 2 (Person B) derives the actual function f. Your job is to set up X, U and the assumptions behind them.

Kinematic, Not Dynamic

There are two kinds of vehicle model:

	Kinematic model (ours)	Dynamic model
Describes	Geometry of motion: positions, angles, velocities	Forces and torques: engine, tyre forces, mass, inertia
Main assumption	Wheels roll without slipping	Tyres slip a little; slip creates the force
Valid when	Low speed, good grip (parking, yards)	Highway speed, hard braking, ice
Inputs	Speed v and steering angle δ	Throttle or brake force and steering torque

At yard speeds (at most 2 m/s, about 7 km/h) tyre slip is a few centimetres, smaller than our 0.25 m safety clearance. So the kinematic model is accurate enough and much simpler. Say this confidently if asked "why not a dynamic model?".

Our Assumptions, in One List

- Motion is **planar**: a flat yard, so each body needs only x, y and a heading angle.
- Tractor and trailer are **rigid bodies**: they do not bend.
- The hitch is a **frictionless pin joint** (revolute joint), placed d = 0.5 m ahead of the tractor's rear axle.
- Wheels **roll without slipping** sideways.
- Each axle's two wheels act like **one wheel at the axle centre** (the "bicycle" simplification, Part 2).
- **One trailer**, and the trailer has one axle (or a tandem group that behaves like one).

4 Two Rigid Bodies and One Passive Joint

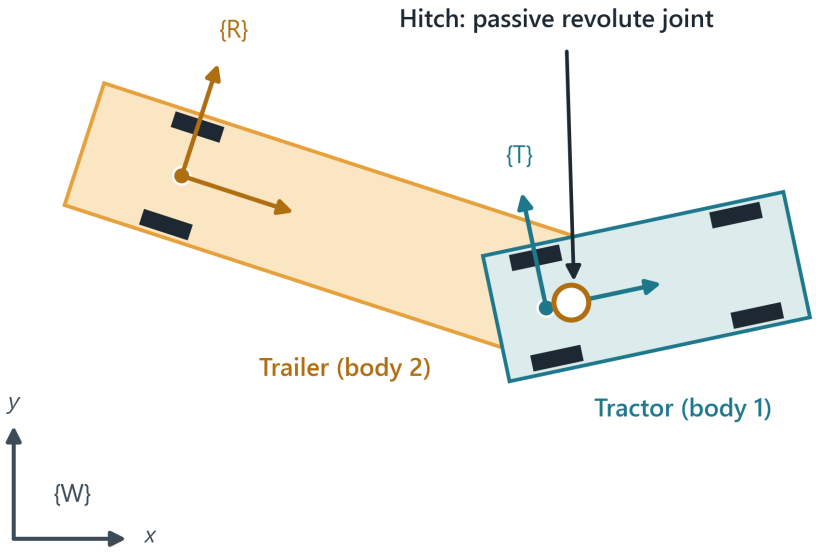


Figure 1. The robot: body 1 (tractor) with frame {T} at its rear axle, body 2 (trailer) with frame {R} at its axle, joined at the hitch. {W} is the fixed world (yard) frame.

Counting Degrees of Freedom

A rigid body moving on a plane needs **3 numbers**: x, y and a heading angle. So two separate bodies would need 6. The hitch pins a point of the tractor to a point of the trailer; that removes **2** freedoms (the two bodies can no longer separate in x or in y), but they can still rotate relative to each other. So:

$$\underbrace{3+3}_{\text{two free bodies}} - \underbrace{2}_{\text{revolute joint}} = 4 \text{ degrees of freedom}$$

This is the planar version of the **Grübler (Kutzbach) count** you saw for robot arms: each revolute joint removes 2 of the 3 planar freedoms.

Like a Two-Link Arm, With Two Differences

	Two-link planar arm	Tractor-trailer
Links	Two rigid links	Two rigid bodies
Joint	Revolute, **driven by a motor**	Revolute, **passive** (no motor at the hitch)
Base	Fixed to the ground	**Moves**: the tractor drives around
Joint angle	q_2	Hitch angle ψ
How the joint angle changes	You command it	Only indirectly, through the wheels (this is why reversing is hard)

SAY IT LIKE THIS

"Think of the rig as a two-link robot arm lying flat on the ground, except the base drives around and the joint has no motor. The only way to change the hitch angle is to drive and steer the tractor."

5 Frames and Rigid-Body Transforms

We attach a coordinate frame to each body: {T} at the tractor's rear-axle centre, pointing along the tractor; {R} at the trailer axle, pointing along the trailer. Knowing the tractor's pose, the rest of the rig follows by chaining transforms, exactly like forward kinematics of an arm.

Unit Vectors

For any heading θ, define the unit vector along it and the one perpendicular to it (rotated 90° anticlockwise):

$$\hat{u}(\theta) = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}, \qquad \hat{n}(\theta) = \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}$$

The Chain: Rear Axle → Hitch → Trailer Axle

1. **Rear axle** (tractor frame origin): $p_1 = (x_1,\ y_1)$.
2. **Hitch**: d metres ahead of the rear axle, along the tractor's heading:
$$p_h = p_1 + d\,\hat{u}(\theta_0)$$
3. **Trailer axle**: L_2 metres behind the hitch, along the trailer's heading:
$$p_2 = p_h - L_2\,\hat{u}(\theta_1), \qquad \theta_1 = \theta_0 - \psi$$

The Same Thing as a Homogeneous Transform

If the professor asks for the matrix form, the tractor's pose in the world is

$$T_{WT} = \begin{pmatrix} \cos\theta_0 & -\sin\theta_0 & x_1 \\ \sin\theta_0 & \cos\theta_0 & y_1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad p_h = T_{WT} \begin{pmatrix} d \\ 0 \\ 1 \end{pmatrix}$$

Then rotate by $-\psi$ at the hitch and move L_2 backwards to reach the trailer axle: $T_{WR} = T_{WT}\,\mathrm{Trans}(d,0)\,\mathrm{Rot}(-\psi)\,\mathrm{Trans}(-L_2,0)$. This is the same chain as a two-link arm: translate, rotate at the joint, translate along the second link.

Worked Example

Take $x_1 = 0$, $y_1 = 0$, $\theta_0 = 15^\circ$, $\psi = 30^\circ$ (the pose in the slide figure). Then $\theta_1 = 15^\circ - 30^\circ = -15^\circ$ and

$p_h = (0.5\cos 15^\circ,\ 0.5\sin 15^\circ) = (0.483,\ 0.129)$ m

$p_2 = p_h - 8\,(\cos(-15^\circ),\ \sin(-15^\circ)) = (0.483 - 7.727,\ 0.129 + 2.071) = (-7.244,\ 2.200)$ m

So the trailer axle sits about 7.2 m behind and 2.2 m to the left of the rear axle. The viewer uses exactly these formulas to draw the trailer, and the planner uses them to check collisions for both bodies.

6 Configuration Space

The **configuration** is the smallest set of numbers that fixes where every point of the robot is. For our rig:

$$q = (x_1,\ y_1,\ \theta_0,\ \theta_1) \in \mathbb{R}^2 \times S^1 \times S^1$$

$\mathbb{R}^2$ is the plane (position), and each S^1 is a circle, because an angle of 360° is the same as 0°. The configuration space is four-dimensional, which matches the 4 degrees of freedom counted in Section 4.

COMMON MISTAKE

Do not say "the configuration has 8 numbers because there are two bodies and four wheels". Wheel rotation angles do not matter for where the truck is, and the hitch removes two numbers. Four is the answer.

7 Rolling Without Slipping: the Nonholonomic Constraint

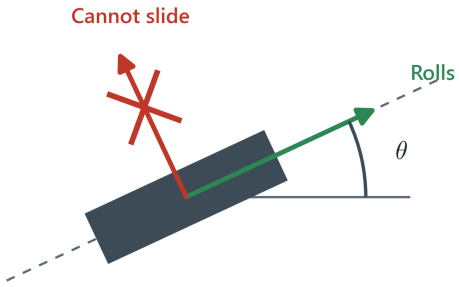


Figure 2. A wheel can roll along its heading $\hat{u}(\theta)$ but cannot slide along $\hat{n}(\theta)$.

Rubber on asphalt grips: a wheel moves forwards or backwards along the direction it points, never sideways. In vector form, the wheel's velocity $\dot{p} = (\dot{x}, \dot{y})$ has **no component along the sideways direction** $\hat{n}(\theta)$:

$$\dot{p} \cdot \hat{n}(\theta) = -\dot{x}\sin\theta + \dot{y}\cos\theta = 0$$

which is usually written

$$\dot{x}\ \sin\theta - \dot{y}\ \cos\theta = 0$$

Where the Constraints Act

- **Tractor rear axle** (heading θ_0): one constraint. This is why the rear axle moves exactly along the tractor's heading, which makes it the natural reference point.
- **Trailer axle** (heading θ_1): one constraint. This is what makes the trailer follow the hitch.
- **Front (steered) axle**: its constraint is what links the steering angle to the turning rate, $\dot{\theta}_0 = v\tan\delta / L_1$. Person B derives it; for you, it is the reason δ appears as an input.

Written together, the constraints have the **Pfaffian form** $A(q)\,\dot{q} = 0$: they restrict the velocities $\dot{q}$, and the restriction depends on where you are, q.

Holonomic or Nonholonomic?

	Holonomic constraint	Nonholonomic constraint
Restricts	Positions: some configurations are impossible	Velocities: some directions of motion are impossible *right now*
Example here	The hitch: the two bodies must meet at one point	The wheels: no sideways velocity
Effect	Removes a dimension from the configuration space (6 → 4)	Does **not** remove a dimension: every pose is still reachable
Can be integrated?	Yes, it is already a position equation	**No**, there is no equation $g(q) = 0$ behind it

Why the Wheel Constraint Cannot Be Integrated

If the no-slip rule could be integrated into a position equation, some positions would be forbidden. They are not: a car can end up one parking space to the side of where it started, with the same heading, without its wheels ever sliding. That is parallel parking:

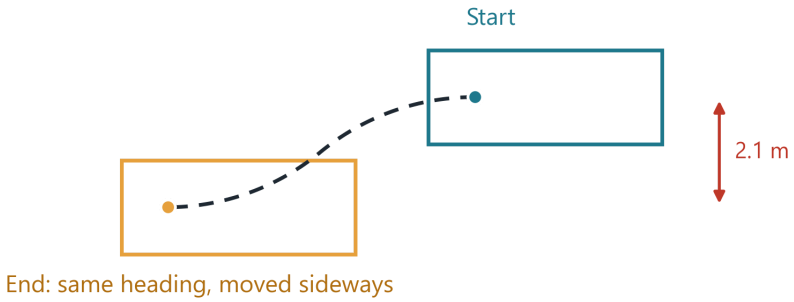


Figure 3. Bicycle model of a car reversing 3.2 m with full right lock, then 3.2 m with full left lock. It ends with the same heading, moved sideways, although its wheels never slid. The constraint limits *how* it moves, not *where* it can get to.

8 Two Inputs, Four States: Still Controllable

Count again: 4 configuration coordinates, minus 2 no-slip constraints (rear axle and trailer axle) leaves **2 directions** in which the rig can move at any instant. Those are exactly our two inputs: drive along the heading (v) and change the heading by steering (δ).

Two inputs, four states: is it still possible to reach every pose? Yes. Switching between the two motions produces motion in new directions, the way "forward-left, reverse-right" in Figure 3 produced sideways motion. Mathematically, combining the two input vector fields (their **Lie bracket**) generates the missing directions, and **Chow's theorem** says that if repeated brackets span all directions, every configuration is reachable. For the car and for the tractor-trailer they do.

SAY IT LIKE THIS

"Four coordinates, two no-slip constraints, so only two free motions at any instant, which are speed and steering. Even so, by combining them the truck can reach any position and heading. That is why parking is possible at all: you cannot slide a truck sideways into a bay, but you can always manoeuvre into it."

COMMON MISTAKE

Do not call the system "not controllable" because it has fewer inputs than states. Fewer inputs means it cannot move in every direction *instantly*; it does not mean it cannot *reach* every configuration.

9 Choosing the State and the Inputs

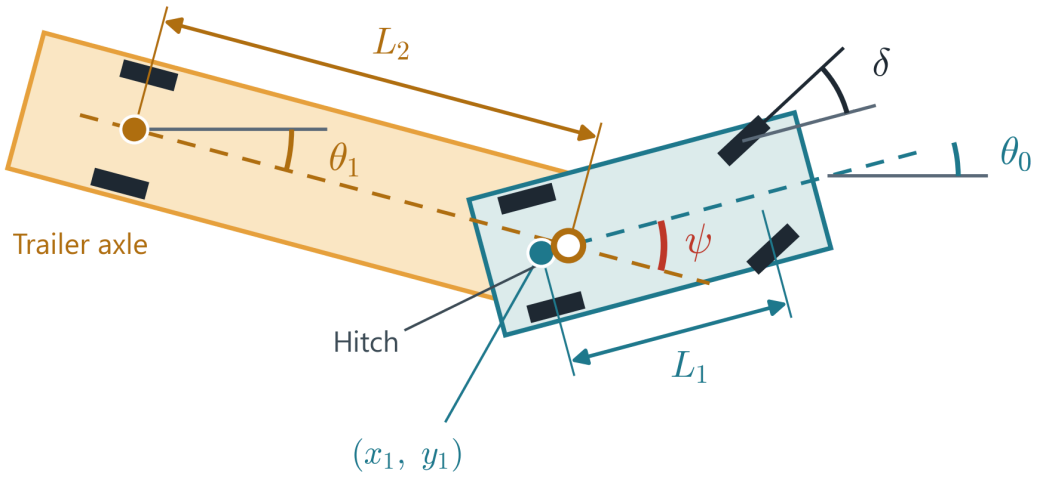


Figure 4. Every symbol of the model. θ_0 and θ_1 are measured from the world x axis; ψ is the angle between the two body axes at the hitch; δ is the front wheels' angle relative to the tractor.

The State Vector

$$X = \begin{pmatrix} x_1 \\ y_1 \\ \theta_0 \\ \psi \end{pmatrix}$$

Symbol	Meaning	Why this choice
$x_1,\ y_1$	Centre of the tractor's rear axle, in the world frame (m)	Its velocity is purely along the heading (no-slip), so its equations are the simplest
θ_0	Tractor heading, from the x axis, anticlockwise positive	Together with x_1, y_1 it fixes the tractor
$\psi = \theta_0 - \theta_1$	Hitch angle: tractor heading minus trailer heading	It is the safety quantity: a jackknife is $\lvert\psi\rvert$ too large, so we limit $\lvert\psi\rvert \le 60^\circ$

Why ψ and not θ_1? They carry the same information ($\theta_1 = \theta_0 - \psi$), so either gives a valid state. We pick ψ because the constraint we care about is on ψ, a real truck measures it with a hitch-angle sensor, and the

instability (Part 2) is an equation in ψ alone.

Sign convention: $\psi > 0$ means the tractor is turned anticlockwise (to the left) relative to the trailer. $\psi = 0$ means the rig is straight.

The Input Vector

$$U = \begin{pmatrix} v \\ \delta \end{pmatrix}$$

Symbol	Meaning	Limit and why
v	Signed speed of the rear axle: positive forward, **negative in reverse** (m/s)	$\lvert v \rvert \leq 2$ m/s: yard speed, where the no-slip assumption holds
δ	Front-wheel steering angle relative to the tractor, positive to the left	$\lvert \delta \rvert \leq 35^\circ$: the steering lock of the truck

These are the two things a driver controls. Our guidance converts δ into steering-wheel turns (a 20 : 1 steering ratio, so full lock is about 1.9 turns) and v into km/h.

Parameters

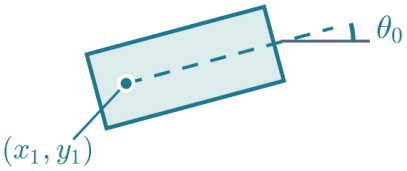
Symbol	Meaning	Value
L_1	Tractor wheelbase (rear axle to front axle)	4.0 m
L_2	Hitch to trailer axle	8.0 m
d	Hitch (fifth wheel) ahead of the tractor's rear axle	0.5 m
$\delta_{\max}$	Steering lock	35°
ψ_{crit}	Hard hitch-angle limit enforced by the controller	60°
ψ_{jack}	Physical jackknife: cab and trailer touch	75°
Width	Both bodies	2.5 m

Why is the hitch ahead of the rear axle? On a real semi-truck the fifth wheel sits slightly ahead of the drive axle so the trailer's weight presses on the driven wheels. It adds a small term to the equations (Part 2).

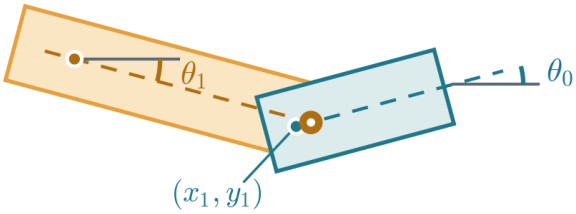
10 How to Draw the Model by Hand

Practise this until you can do it in about a minute on a whiteboard. Draw in this order, so every angle has a clear reference line. Exaggerate the angles (15° to 30°) so the arcs are easy to see.

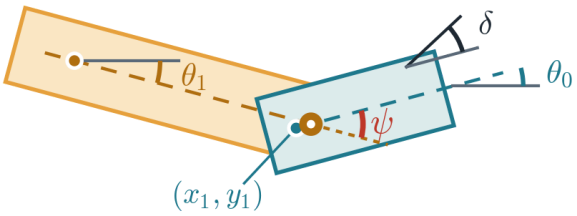
1 Tractor, rear axle, heading



2 Hitch, trailer, trailer heading



3 Hitch angle and steering



4 Lengths

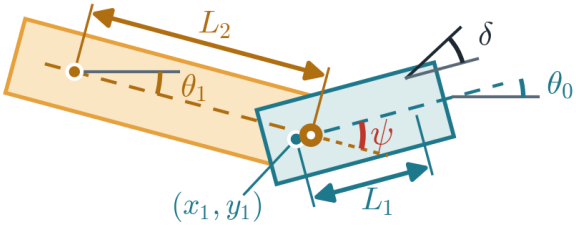


Figure 5. The drawing built up in four stages. Copy this order.

1. **Tractor.** Draw a rectangle tilted up about 15°. Put a dot near its back for the rear-axle centre and label it (x_1, y_1). Draw the tractor's centre line dashed, and extend it past the front. Ahead of the tractor, draw a short horizontal line and mark the angle up to the centre line: that is θ_0.
2. **Hitch and trailer.** Mark the hitch as a small circle just ahead of the rear axle (a short distance d). From the hitch, draw the trailer as a longer rectangle going backwards, tilted the other way (about −15°). Put a dot at the trailer axle near its back end. From that dot draw a horizontal line and mark the angle to the trailer's centre line: that is θ_1.
3. **Hitch angle and steering.** Extend the trailer's centre line forward through the hitch (dotted). The angle at the hitch between this extension and the tractor's centre line is ψ. At the front axle, draw a wheel turned relative to the tractor; the angle between the tractor's axis and the wheel is δ.
4. **Lengths and speed.** Draw a dimension arrow from the rear axle to the front axle (L_1) and from the hitch to the trailer axle (L_2). Draw an arrow along the tractor for v (say "negative when reversing").

COMMON MISTAKES WHEN DRAWING

- Measuring ψ from the trailer's centre line *behind* the hitch. ψ is between the tractor's axis and the trailer's axis *extended forward through the hitch*.
- Putting the hitch at the tractor's back bumper. It is just ahead of the rear axle (d = 0.5 m).
- Measuring δ from the world x axis. δ is relative to the tractor.
- Measuring L_2 from the trailer's front edge. L_2 runs from the hitch to the trailer axle.
- Drawing θ_0 and ψ arcs on top of each other at the rear axle. Draw θ_0 ahead of the tractor (the heading is the same anywhere along the body).

QUICK CHECK AFTER DRAWING

Point at your drawing and say: "θ_0 is 15°, θ_1 is -15°, so $\psi = \theta_0 - \theta_1 = 30^\circ$." If the three angles you drew are consistent with this sum, the drawing is right.

11 What to Say on Each Slide

Slide 5 · Part 1 divider (10 s)

"In Part 1 we model the tractor-trailer: what kind of robot it is, which laws govern it, and how we describe its state."

Slide 6 · A Tractor-Trailer Is a Wheeled Mobile Robot (about 1 min)

Two rigid bodies, frames {T} and {R} in the world frame {W}. One revolute joint with no motor. Configuration needs four numbers (count 3 + 3 − 2). Only two inputs are driven: under-actuated, the trailer just follows.

Slide 7 · Three Robotics Principles Behind the Model (about 1.5 min)

Rigid-body transforms (the hitch and trailer follow from the tractor pose, like a two-link arm). Rolling without slipping (no sideways velocity; write the constraint). Nonholonomic but controllable (cannot move sideways, can reach any pose: parallel parking, Chow's theorem).

Slide 8 · State Variables and Control Inputs (about 1.5 min)

Point at each symbol in the figure: x_1, y_1, θ_0, ψ; then v (negative in reverse) and δ. Explain why ψ and not θ_1. Read the parameters. Hand over: "Now Person B will show how these four numbers change when the driver steers and drives."

12 Questions the Professor May Ask

Why is the system called nonholonomic?

Because its constraints restrict velocities, not positions, and cannot be integrated into position equations. No sideways velocity at each axle limits how it moves, but every pose remains reachable.

How many degrees of freedom does the rig have?

Four configuration coordinates: 3 + 3 for two planar bodies minus 2 for the revolute joint. At any instant only 2 of them can change freely, because there are 2 no-slip constraints; those 2 are the inputs.

Why did you choose the rear axle as the reference point?

The no-slip constraint makes its velocity point exactly along the heading, so $\dot{x}_1 = v\cos\theta_0$ and $\dot{y}_1 = v\sin\theta_0$, the simplest possible equations. Any other point would add terms in $\dot{\theta}_0$.

Why ψ instead of the trailer heading θ_1?

Same information, since $\theta_1 = \theta_0 - \psi$. But the safety limit is on ψ, the instability is an equation in ψ, and a real truck measures ψ directly.

Is the system controllable with only two inputs?

Yes. The Lie brackets of the two input vector fields span all four directions, so by Chow's theorem every configuration is reachable. Parallel parking is the everyday example.

Why not a dynamic model?

At yard speeds (≤ 2 m/s) tyre slip is a few centimetres, well inside our 0.25 m clearance margin. The kinematic model is accurate there and needs no tyre or mass data.

What is the difference between the hitch constraint and the wheel constraint?

The hitch constraint is holonomic: the two bodies must share one point, which removes two coordinates. The wheel constraints are nonholonomic: they only forbid sideways velocity.

What is the configuration space?

$\mathbb{R}^2 \times S^1 \times S^1$: a position in the plane and two angles, each living on a circle.

What happens with two trailers?

Three bodies, two joints: 9 − 4 = 5 coordinates; three axles give three no-slip constraints, leaving 2 free motions. Still the same two inputs, and still controllable, but with one more unstable hitch.

13 Check Yourself

1. If $\theta_0 = 40^\circ$ and $\theta_1 = 25^\circ$, what is ψ? Is the tractor turned left or right relative to the trailer?
2. The rear axle is at (10, 5) m, $\theta_0 = 90^\circ$ and $\psi = -20^\circ$. Find θ_1, the hitch position and the trailer-axle position.
3. Count the configuration coordinates and the instantaneous free motions of a car (one body, front-wheel steering).
4. Write the no-slip constraint for the trailer axle, whose position is (x_2, y_2).
5. True or false: a nonholonomic robot cannot reach some positions.
6. Why does a revolute joint remove two freedoms, not one?

Answers

1. $\psi = 40^\circ - 25^\circ = 15^\circ$. Positive, so the tractor is turned to the left (anticlockwise) relative to the trailer.
2. $\theta_1 = \theta_0 - \psi = 90^\circ + 20^\circ = 110^\circ$. Hitch: $(10 + 0.5\cos 90^\circ,\ 5 + 0.5\sin 90^\circ) = (10,\ 5.5)$ m. Trailer axle: $(10 - 8\cos 110^\circ,\ 5.5 - 8\sin 110^\circ) = (12.74,\ -2.02)$ m.
3. 3 coordinates (x, y, θ), 1 no-slip constraint on the rear axle, so 2 free motions: drive and steer.
4. $\dot{x}_2 \sin\theta_1 - \dot{y}_2 \cos\theta_1 = 0$.
5. False. It can reach every position; it just cannot move in every direction at every instant.
6. A pin forces two points to coincide, which is two equations (x and y equal). Rotation about the pin stays free.

14 One-Page Cheat Sheet

The Robot

Two rigid bodies + one passive revolute joint. DOF: $3 + 3 - 2 = 4$. Configuration $q = (x_1, y_1, \theta_0, \theta_1) \in \mathbb{R}^2 \times S^1 \times S^1$.

Transforms

$\hat{u}(\theta) = (\cos\theta, \sin\theta)$, $\hat{n}(\theta) = (-\sin\theta, \cos\theta)$. Hitch $p_h = p_1 + d\,\hat{u}(\theta_0)$. Trailer axle $p_2 = p_h - L_2\,\hat{u}(\theta_1)$, $\theta_1 = \theta_0 - \psi$.

No-Slip (Nonholonomic)

$\dot{x}\sin\theta - \dot{y}\cos\theta = 0$ at the rear axle and at the trailer axle. Velocity constraint, not integrable; pose still reachable (parallel parking, Chow's theorem).

State and Inputs

$X = [x_1, y_1, \theta_0, \psi]^T$, $U = [v, \delta]^T$. $v < 0$ in reverse. $\psi > 0$: tractor turned left of the trailer.

Numbers

L_1 = 4 m, L_2 = 8 m, d = 0.5 m, $|\delta| \le 35^\circ$, $|\psi| \le 60^\circ$ (jackknife at 75°), $|v| \le 2$ m/s.

Three Sentences to Remember

(1) It is a wheeled robot: two bodies, one passive joint. (2) Wheels cannot slide, so it is nonholonomic but still reaches every pose. (3) Four states, two inputs; ψ is a state because it is what we must keep below 60°.